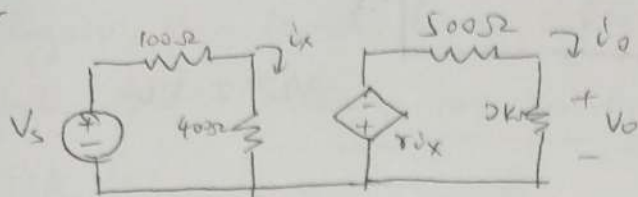


HW # 5 Soln.

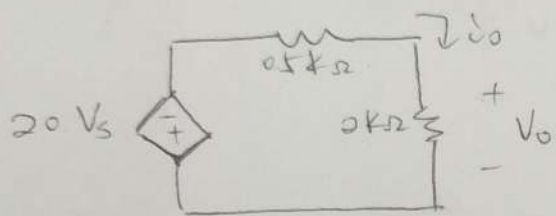
4.1



$$V = 10 \text{ V}$$

$$i_x = \frac{V_s}{100\Omega + 400\Omega} = \frac{V_s}{500\Omega} = 2 \text{ V}_s \text{ mA}$$

then : rewrite the circuits

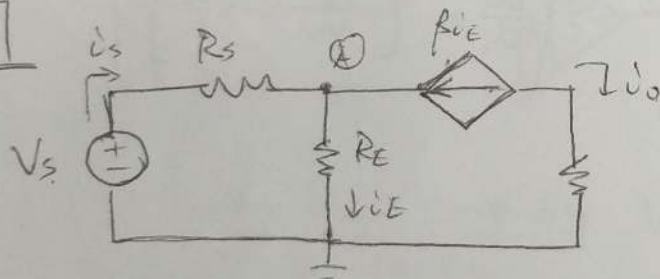


$$i_o = \frac{20 V_s}{2.5 K\Omega} = 4 i_x \Rightarrow \boxed{\frac{i_o}{i_x} = 4}$$

$$V_o = \frac{2 K\Omega}{2 K\Omega + 0.5 K\Omega} \cdot 20 \cdot V_s$$

$$\Rightarrow \boxed{\frac{V_o}{V_s} = 16}$$

4.7



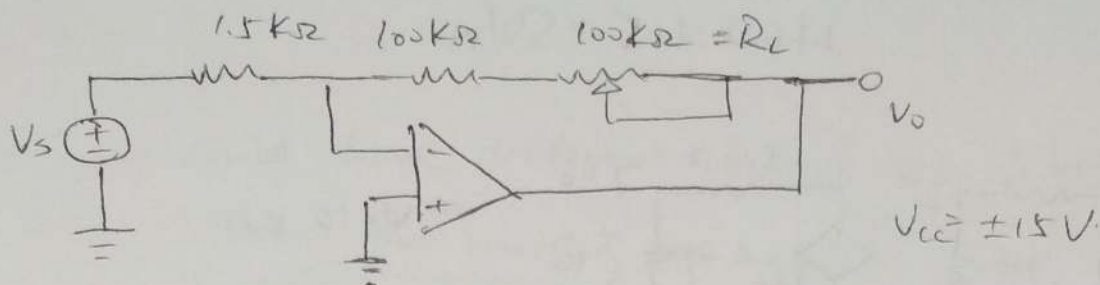
$$i_s = \frac{V_s - V_A}{R_s}, \quad i_o = -\beta i_E, \quad i_E = \frac{V_A - 0}{R_E} = \frac{V_A}{R_E} = -\frac{1}{\beta} i_o$$

At node A : KCL

$$i_s = i_o + i_E \Rightarrow i_s = i_o - \frac{1}{\beta} i_o$$

$$\Rightarrow \boxed{\frac{i_o}{i_s} = \frac{1}{1 - \frac{1}{\beta}}}$$

4.31



$$\frac{V_o}{V_s} = - \frac{100 k\Omega + R_L}{1.5 k\Omega} \quad \text{when } R_L = 0, \quad \frac{V_o}{V_s} = - \frac{200}{3}, \quad \therefore V_o = - \frac{200}{3} V_s$$

$$\text{when } R_L = 100 k\Omega, \quad \frac{V_o}{V_s} = - \frac{400}{3}, \quad \Rightarrow V_o = - \frac{400}{3} V_s$$

$$\text{Since, } V_{cc} = \pm 15 V, \quad \Rightarrow -15 V < V_o < +15 V.$$

$$\text{if } - \frac{200}{3} V_s \leq -15, \quad \text{then,}$$

$$\boxed{V_o = -15 V}$$

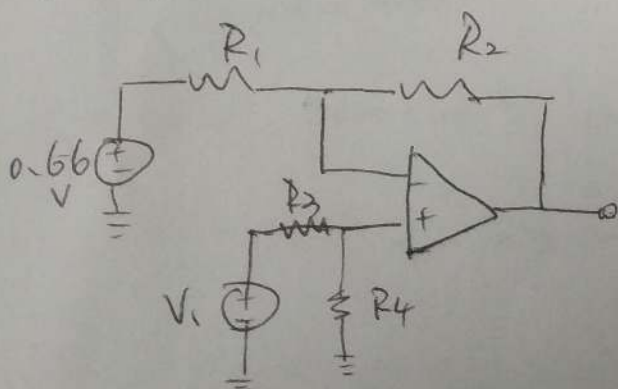
$$\text{if } - \frac{400}{3} V_s \leq -15 \leq - \frac{200}{3} V_s, \quad \text{then,}$$

$$V_o \in [-15, - \frac{200}{3} V_s] \quad \Rightarrow \quad \boxed{\frac{V_o}{V_s} \in \left[-\frac{15}{V_s}, -\frac{200}{3} \right]}$$

$$\text{if } -15 \leq - \frac{400}{3} V_s, \quad \text{then,}$$

$$\boxed{\frac{V_o}{V_s} \in \left[-\frac{400}{3}, -\frac{200}{3} \right]}$$

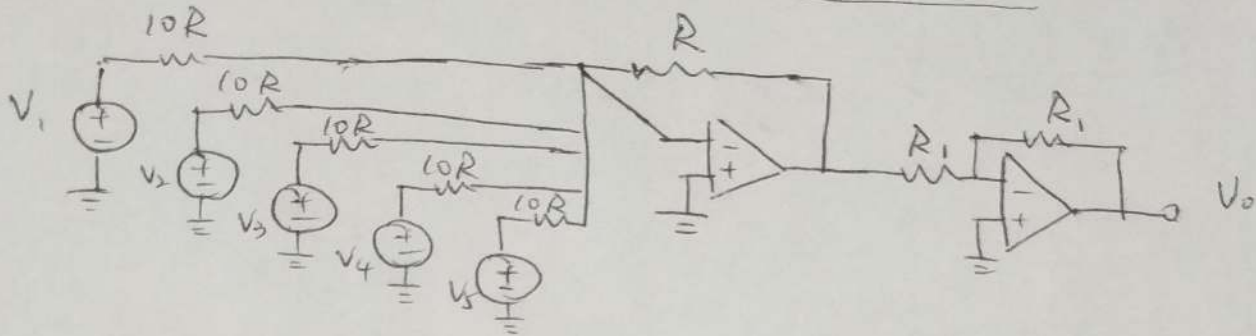
4.33 There are different possible solutions, here is only one:



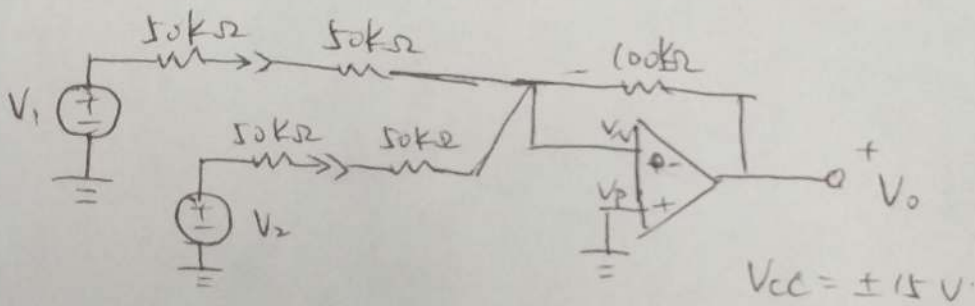
$$\frac{R_3}{R_4} = \frac{1}{5} = \frac{R_1}{R_2}$$

4.40

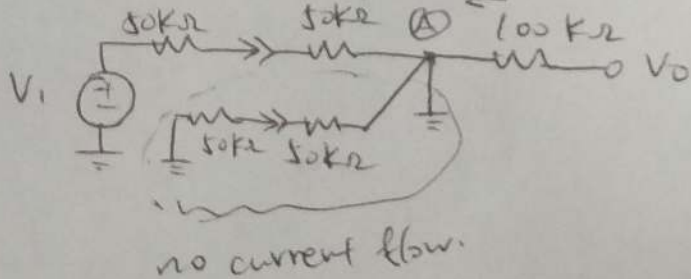
Again, you could have different design. Here is a simple one: Using two inverting Op-Amp



4.41



a) Zero out V_2 , consider V_1 , apply $V_p = V_n$



KCL at node A :

$$\frac{V_1}{50k + 50k} + \frac{V_0}{100k} = 0$$

$$\Rightarrow \frac{V_0}{V_1} = -1$$

Similarly, zero out V_1 , Consider V_2

$$\frac{V_0}{V_2} = -1$$

→ thus,
$$V_0 = -V_1 - V_2$$

b), To prevent from saturating, $|V_o| \leq 15V$

$$\text{if } v_1 = 1V, \Rightarrow V_o = -1 - V_2 = -(1 + V_2)$$

$$|-(1 + V_2)| \leq 15V \Rightarrow -15V \leq 1 + V_2 \leq 15V$$

$$\Rightarrow \boxed{-16V \leq V_2 \leq 14V}$$